Qn14. If we start with 2 and then apply the instructions as given in steps 1 to 5 we have the following.

STEP	
1	$2+4=6$
2	$6-1=5<15$
3	$5+4=9$
2	$9-1=8<15$
3	$8+4=12$
2	$12-1=11<15$
3	$11+4=15$
2	$15-1=14<15$
3	$14+4=18$
2	$18-1=17>15$
4	$17+3=20>18$
5	$20-2=18$

Therefore the card result is 18.

Qn15. (a) Let the length of $AB=AC=CF=x$.
Perimeter of the figure is $x+x+x+x+25+x+x-25=180 \Longrightarrow 6x=180 \Longrightarrow x=30$. Therefore the length $FE=x-25=30-25=5$ cm.

(b) Area of the figure is $x^2\times(x\times 25)=30^2+(30\times 25)=900+750=1650\text{cm}^2$.

NMC 2016 PRIMARY LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. In the text line of numbers below, insert "+" between the numbers as many times as necessary so that 987654321 = 90.

Qn2. Evaluate

$$\frac{3^3+3\times 3+3}{3^2+3^1+3^0}$$

Qn3. What is the value of $\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}$?

Qn4. At a farm, one plant is currently 1 meter tall and grows at a rate of 6cm per week. A second plant is currently 0.7 meters tall and grows at a rate of 9cm per week. In how many weeks will both plants be of the same height?

Qn5. Kahwa thinks of a whole number less than 30. When he divides the number by 6 the remainder is 3. When he divides the number by 8 the remainder is 5. What is the number?

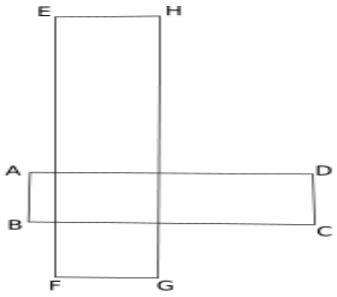
Qn6. A sale advert read: "Buy three mathematics books at the regular price and get the fourth book at Shs. 3000." Kato paid Shs. 24000 for the set of four books at the sale. What was the regular price of one book?

Qn7. It is 215 minutes to midday. What time is it on a 24-hour digital clock?

Qn8. Each game in a knock-out tournament involves 7 players and only the winner of the game remains in the tournament (the other players are knocked out). The tournament was played in 8 games to declare the final winner. How many players were there at the start of the tournament?

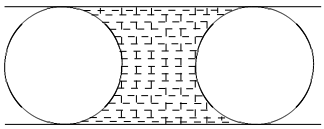
Qn9. Babirye earns Shs. $300,000$ per week for a 44 hour week. Her weekly salary is increased by 8% and her hours are reduced by a fifth. How much does she earn now per hour?

Qn10. In the figure shown, the overlap of rectangles ABCD and EFGH is also a rectangle. The area of ABCD is $21cm^2$. The area of EFGH is $50cm^2$. $AB=3cm$ and $EH=5cm$. Find the area of the entire figure.



SECTION B

Qn11. Two circles, each of diameter 28cm, fit exactly between two parallel lines as shown in the diagram below. The centres of the circles are 35cm apart. What is the area of the shaded part? (Use $\pi = \frac{22}{7}$).



Qn12. A digital clock shows 2:35. This is the first time after midnight when all three digits are different prime numbers.

(a) At how many times between midnight and midday will the digits on the clock be three different prime numbers?

(b) What is the last time before midday when all three digits are three different prime numbers?

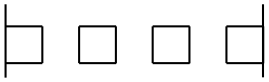
Qn13. Mutesi and Akello played against each other in a game of 12 rounds. In each round, the winner scored 5 points and the loser scored 3 points. At the end of the game, Akello's total score was 44 points. How many points did the winner attain and who won the game?

Qn14. Some fractions can be written as a sum of different unit fractions (i.e. a fraction with a 1 as the numerator). For example, $\frac{23}{24} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{12}$.

(a) Write $\frac{17}{18}$ as a sum of any number of different unit fractions.

(b) Write 1 as the sum of exactly three different unit fractions.

Qn15. Norah has four pictures in a square shape to hang in a horizontal row on a wall which is 2.2 meters wide. The pictures are identical in size and are 40cm wide. She hangs the first two pictures so that one is on the extreme left of the wall and the other is on the extreme right of the wall. She hangs the remaining two pictures in such a way that all the four pictures are equally spaced. What is the distance between the mid-points of any two pictures?



NMC 2016 PRIMARY LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1. $9 + 8 + 7 + 6 + 54 + 3 + 2 + 1 = 90$.

Qn2. $\frac{3^3+3\times3+3}{3^2+3^1+3^0} = \frac{27+9+3}{9+3+1} = \frac{39}{13} = 3$.

Qn3. $\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{4+2+1}{8} = \frac{7}{8}$.

Qn4. Let x be the duration in weeks that will have elapsed for both plants to be of the same height.

We know that $1m = 100cm \implies 0.7m = 70cm$. So, $100 - 6x = 70 + 9x \implies 9x - 6x = 100 - 70 = 30$. Hence $3x = 30 \implies x = 10$. Therefore after 10 weeks, the two plants will have the same height.

Qn5. The numbers that are less than 30 that leave remainder 3 when divided by 6 are $3, 9, 15, 21, 27$. Similarly the numbers that are less than 30 that leave remainder 5 when divided by 8 are $5, 13, 21, 29$. The common number in both lists is 21. Therefore the number is 21.

Qn6. Let the regular price of each of 3 books be p. The 4th book cost Shs. 3000. Since Kato bought a set of 4 books, it implies that he bought the first 3 books each at p and the fourth one at Shs. 3000. Therefore $3p + 3000 = 24000 \Longrightarrow 3p = 21000 \Longrightarrow p = \frac{21000}{3} = 7000$. The regular price of one book is Shs.7000

Qn7. 215 minutes is equal to $\frac{215}{60} = 3$hrs 35min. So, $12:00 - 3:35 = 8:25$. Therefore 3hrs 35min to midday implies that the time is 0825hrs.

Qn8. Since each game involves 7 players, each game involves 6 losers and 1 winner. 8 games have $6 \times 8 = 48$ losers. There is one eventual winner. Therefore the number of players at the start of the tournament is $48 + 1 = 49$.

Alternatively
The first game has 7 players. Each of the next 7 games has 6 new players. Therefore the total number of players in the tournament is $7 + (6 \times 7) = 7 + 42 = 49$.

Qn9. Babirye's new weekly pay is $= 300000 \times \frac{108}{100} = 324,000$.
New weekly hours $= 44 \times \frac{4}{5} = 35.2$ hours.
New weekly rate $= \frac{324000}{35.2} = 9204.54$. Thus Babirye's new earning is Shs.9204.54 per hour.

Qn10. The area of the entire figure = Area of $ABCD$ + Area of $EFGH$- Area of $PQRS$.
The area of $PQRS$ is counted twice because of the overlap.
Area of $PQRS = \overline{PQ} \times \overline{PS} = 3 \times 5 = 15\text{cm}^2$. Therefore the area of the entire figure $= (21 + 50 - 15) = 56\text{cm}^2$.

SECTION B

Qn11.

$$\begin{aligned}
\text{Area of shaded part} &= \text{Area of ABC} - 2 \times \text{Area of semicircle.} \\
&= \text{Area of ABCD- Area of the circle.} \\
&= (28 \times 35) - \left(\frac{22}{7} \times \frac{28}{2} \times \frac{28}{2}\right) \\
&= (28 \times 35) - (22 \times 28) \\
&= 28(35 - 22) = 28 \times 13 = 364cm^2.
\end{aligned}$$

Qn12. (a) The single digit prime numbers are $2, 3, 5, 7$. The times are $2:35, 2:37, 2:53, 2:57, 3:25, 3:27, 3:52, 3:57, 5:23, 5:27, 5:32, 5:37, 7:23, 7:32, 7:35, 7:52, 7:53$. Therefore there are 18 times between midnight and midday when the three digits are different numbers.

(b) The last time is 7 : 53

Qn13. Let x be the total number of the games won and y be the total number of the games lost. For Akello; $5x + 3y = 44$ and $x + y = n$ where n is the number of rounds. Substituting the second equation into the first we get

$$\begin{aligned}
5x + 3(n - x) &= 44 \\
5x + 3n - 3x &= 44 \\
x &= \frac{44 - 3n}{2} = 22 - \frac{3n}{2}. \\
y &= n - (22 - \frac{3n}{2}) = \frac{5n}{2} - 22.
\end{aligned}$$

For the winner, $x > y$ meaning

$$x = 22 - \frac{3 \times 12}{2} = 22 - 18 = 4$$

and

$$y = \frac{5n}{2} - 22 = \frac{5 \times 12}{2} - 22 = 30 - 22 = 8.$$

Hence Akello won 4 games implying that Mutesi won the game with $12 - 4 = 8$ wins.
In total Mutesi won the game with $5 \times 8 + 3 \times 4 = 40 + 12 = 52$ points.

Qn14. (a) Identify the divisors of 18 so that their sum is 17. Then identify the corresponding factors of 18 which when multiplied by the divisor gives 18. The divisors are $18, 9, 6, 3, 2$ whose corresponding factors of 18 are $1, 2, 3, 6, 9$.
sum the quotients of the corresponding factor. Divisors summing to 17 are $9, 6, 2$ that is $9+6+2 = 17$. Then corresponding factors are $2, 3, 9$ respectively. Then the sum of these quotients are

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{9} = \frac{9+6+2}{18} = \frac{17}{18}.$$

(b) Identify three numbers whose sum is equal to their LCM. These are $1, 2, 3$. The divisors are $1, 2, 3$ and the corresponding factors are $6, 3, 2$. The sum of these quotients are $\frac{1}{6} + \frac{1}{3} + \frac{1}{2} = \frac{1+2+3}{6} = \frac{6}{6} = 1$. Therefore $1 = \frac{1}{6} + \frac{1}{3} + \frac{1}{2}$.

Qn15. The distance between extreme pictures is $2.2 - (2 \times 0.4) = 2.2 - 0.8 = 1.4m$. Let the distance between two midpoints of two pictures be x. There are 3 such distances and two half pictures, that is 2×0.2 at the extreme ends. Thus $3x + 2 \times 0.2 = 2.2 \Longrightarrow 3x = 1.8$. Thus $x = 0.6m = 60cm$. Therefore the distance between the mid-points of any two pictures is $60m$.

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NMC 2016 0-LEVEL PAPER 1 QUESTIONS

SECTION A

Qn1. The reciprocal of $1 - \frac{1}{x}$ is -2. What is the value of x.

Qn2. Let $A + R = 17, R + M = 1$, and $M + L = 24$. Compute $A + L$.

Qn3. Suppose $(6^{30} + 6^{-30})(6^{30} - 6^{-30}) = 3^x 8^y - 3^{-x} 8^{-y}$. Find the value of x and y.

Qn4. Given that $(24_b)^2 = 642_b$ where $b > 1$. Find the value of b.

Qn5. Trickster Rabbit agrees with Foolish Fox to double Fox's money every time Fox crosses the bridge nearby Rabbit's house, as long as Fox pays Shs.400 in toll to Rabbit after each crossing. The payment is made after the doubling, Fox is excited about his good fortune until he discovers that all his money is gone after crossing the bridge three times. How much money did Fox have at the beginning?

Qn6. Charles was born in the twentieth century. On his birthday in the year 2014, he noticed that his age was twice the number formed by the rightmost two digits of the year in which he was born. Compute the four-digit year in which Charles was born.

Qn7. Given that

$$\frac{a}{b+d+c} = \frac{4}{3} \text{ and } \frac{a}{b+c} = \frac{3}{5}.$$

Find the value of $\frac{d}{a}$.

Qn8. Peter, Sofia, and Mary got some eggs at a party. Peter had three times as many eggs as Sofia, and Sofia had twice as many eggs as Mary. Peter decides to give some of his eggs to Sofia and Mary so that all three will have the same number of eggs. What fraction of his eggs should Peter give to Sofia?

Qn9. Denis, Henry, Jack, Martha, Ronald, and Tom were in a 12-person race with 6 other people. Ronald finished 6 places ahead of Henry. Martha finished 1 place behind Jack. Denis finished 2 places behind Henry. Jack finished 2 places behind Tom. Tom finished 1 place behind Ronald. Martha finished in 6th place. Who finished in 8th place?

Qn10. At Mulago Hospital one year, multiple birth statistics were as follows: Sets of twins, triplets, and quadruplets accounted for 1000 of the babies born. There were four times as many sets of triplets as sets of quadruplets, and there was three times as many sets of twins as sets of triplets. How many of these 1000 babies were in sets of triplets?

SECTION B

Qn11. The sum of the lengths of the three sides of a right angled triangle is 56. The sum of the squares of the lengths of the three sides of the same right angled triangle is 1250. What is the area of the triangle?

Qn12. A circle is inscribed in a right angled triangle measuring $3cm, 4cm$ and $5cm$, Without use of construction method, find the circumference of the circle?

Qn13. The four numbers $a < b < c < d$ can be paired in six different ways. If each pair has a different sum and if the four smallest sums are $1, 2, 3$ and 4. hat are all possible values of D.

Qn14. ABC ia a right angled triangle. Points D, E and F are on the sides $\overline{AB}, \overline{AC}$ and $\overline{BC}$ respectively such that $\overline{AD} = \overline{AE}$ and $\overline{CE} = \overline{CF}$. Find angle DEF.

Qn15. (a) Prove that if three consecutive integers are multiplied and the middle number is added to the result, the number obtained will always be a cube.

(b) Brenda's school is on a semester system, with three terms per year. The grade point average(GPA) for a student is the sum of the percentage grades (maximum 100%) in each course divided by the total number of courses. The first term she took five courses, and her GPA was 75%. The second term she took four courses, and her GPA was 80%, in the third term, she took three courses. In one of these courses, her grade was 90%. If her overall GPA for the full year is 80%, what is the lowest possible grade she could have obtained in one of the other two courses she took in the third term?

NMC 2016 0-LEVEL PAPER 1 SOLUTIONS

SECTION A

Qn1.

$$\frac{1}{1-\frac{1}{x}} = -2 \Longrightarrow \frac{x}{x-1} = -2$$

$$x = -2x + 2 \Longrightarrow x = \frac{2}{3}.$$

Qn2. Adding the three equations gives

$$\begin{aligned} A+R+M+L+R+M &= 17+24+1=42 \\ A+L+2(R+M) &= 42 \\ A+L &= 42-2=40. \end{aligned}$$

Alternatively; Note that $A+L=(A+R)+(M+L)-(R+M)=17+24-1=40$.

Qn3. By using difference of two squares we have $(6^{30})^2-(6^{-30})^2=3^x8^y-3^{-x}8^{-y}$
$6^{60}-6^{-60}=3^x8^y-3^{-x}8^{-y}$ This means that $6^{60}=3^x8^y=3^x2^{3y}$ or $2^{60}3^{60}=3^x2^{3y}$. Clearly $x=60$ and $3y=60\Longrightarrow y=20$.

Qn4. Expressing in base ten, we have $(2b+4)^2=6b^2+4b+2$. Now

$$\begin{aligned} 4b^2+16b+16 &= 6b^2+4b+2 \\ b-6b-7 &= 0 \\ (b-7)(b+1) &= 0 \\ b &= 7 \text{ or } -1. \end{aligned}$$

Since $b>1$, we conclude that $b=7$.

Qn5. Let x be the amount that the Fox has before crossing for the third time. Then we have that $2x-400=0\Longrightarrow x=200$. This means Fox remained with 200 at the end of the second crossing. Then if y is the amount that remained before the second crossing, then $2y-400=200\Longrightarrow y=300$. Then if z is the initial amount Fox had before the first crossing, then $2z-400=300\Longrightarrow z=350$. Therefore Fox initially had Shs. 350.

Qn6. Let N be the number formed by the rightmost two digits of the year in which Charlie was born. Then his current age is $100-N+14=114-N$. Setting this equal to $2N$ and solving yields $N=38$, hence the answer is 1938.

Alternatively
Let N be the number formed by the rightmost two digits of the year in which Charlie was born. The number of years from 1900 to 2014 can be thought of as the number of years before Charlie was born plus the number of years since he was born, or N plus Charlie's age. Thus $N+2N=114$, which leads to $N=38$, so the answer is 1938.

Qn7. $\frac{a}{b+d+c}=\frac{4}{3}$ and $\frac{a}{b+c}=\frac{3}{5}\Longrightarrow b+c=\frac{5a}{3}$. Substituting this equation into the first we get

$$3a=4(d+\frac{5a}{3})=4d+\frac{20a}{3}$$

$$\frac{-11a}{3}=4d\Longrightarrow\frac{d}{a}=\frac{-11}{12}.$$

Qn8. Assign a variable to the number of eggs Mary has, say m. Then, because we are given that Sofia has twice the number of eggs Mary has, Sofia has $2m$ eggs, and Peter, having three times the number of eggs as Sofia, has $6m$ eggs.

For them to all have the same number of eggs, they must each have $\frac{m+2m+6m}{3}=3m$ eggs. This means Peter must give $2m$ eggs to Mary and m eggs to Sofia, so the answer is $\frac{m}{6m}=\frac{1}{6}$.

Qn9. Because Martha was 6th, Jack was 5th, so Tom was 3rd. Thus, Ronald was 2nd and the 8th place finisher was Henry.

Qn10. We can set up a system of equations where a is the sets of twins, b is the sets of triplets, and c is the sets of quadruplets.

$$\begin{aligned} 2a+3b+4c &= 1000 \\ b &= 4c \\ a &= 3b \end{aligned}$$

Solving for c and a in the second and third equations and substituting into the first equation yields

$$\begin{aligned} 2(3b)+3b+4(0.25b) &= 1000 \\ 6b+3b+b &= 1000 \\ b &= 100 \end{aligned}$$

Since we are trying to find the number of babies and NOT the number of sets of triplets, the solution $4c$. Therefore, we strategically use the second initial equation to realize that $b = 4c$, leaving us with the number of babies born as triplets equal to 100.

SECTION B

Qn11. Let the lengths be a, b, c. Then we have the following equations.

$$\begin{aligned} a+b+c &= 56 \\ a^2+b^2+c^2 &= 1250 \\ a^2+b^2 &= c^2 \end{aligned}$$

Combining the second and third equations, we get $2c^2 = 1250 \implies c = 25$. Hence $a+b+25 = 56 \implies a+b = 31$. This put in the second equation gives $a^2+b^2 = 1250-625 = 650$.
From the identity $(a+b)^2 = a^2+b^2+2ab$ we have $2ab = (a+b)^2-(a^2+b^2) = 31^2-25^2 = 336 \implies ab = 168$. Therefore the area of a triangle $A = \frac{1}{2}ab = \frac{1}{2} \times 168 = 84$ sq. units.

Qn12. Consider Fig 1 and we draw the lines from the incentre to the points of tangency, we know that these meet at right angles so that the square of side c is formed at the right angle. Hence $c = r$ where r is the radius of the circle. But

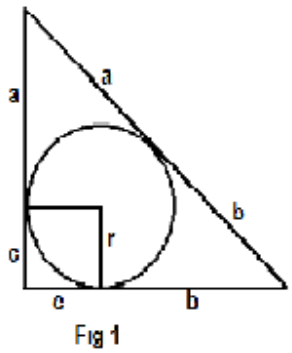


Fig 1

or

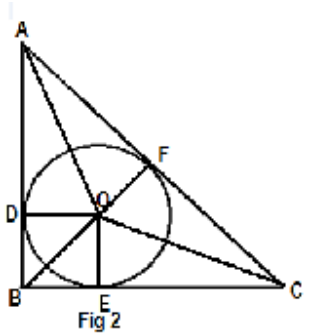


Fig 2

$$\begin{aligned} a+b &= 5 \\ b+c &= 4 \\ a+c &= 3. \end{aligned}$$

Subtracting the third equation from the second equation we get $b-a = 1$. This equation added to the first equation gives $2b = 6 \implies b = 3$. It follows that $a = 2$ and $c = 1$. The radius of the circle is 1 and the circumference is thus $2\pi r = 2\pi \times 1 = 2\pi$.

Alternatively by considering Fig 2

We draw the lines OA, OB, OC from the incentre O to the points A, B, C and the perpendiculars OD, OE, OF from the incentre to the D, E, F of tangency.

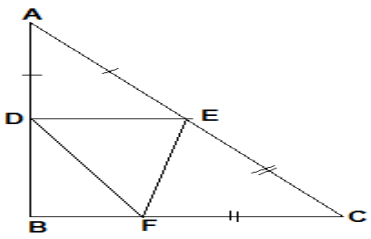
Let the lengths of these perpendiculars be r which is the radius of the circle.
Area of $\triangle ABC$= Area of $\triangle BOC$ + Area of $\triangle AOC$ + Area of $\triangle AOB$. The lines OD, OE, OF are the heights of these triangles, hence

$$\frac{1}{2} \times 3 \times 4 = \frac{1}{2} \times 4 \times r + \frac{1}{2} \times 3 \times r + \frac{1}{2} \times 5 \times r$$

Hence $6 = 2r + 1.5r + 2.5r \Longrightarrow 6 = 6r \Longrightarrow r = 1$.
Hence the circumference of the circle is $2\pi r = 2\pi \times 1 = 2\pi$.

Qn13. The six possible sums are $a+b, a+c, a+d, b+c, b+d$ and $c+d$. Of these the smallest sums are $a+b=1$ and $a+c=2$. These two equations combined give $c = 1+b$. The largest possible sums are $c+d$ and $b+d$. The sums in the middle are therefore $b+c$ and $a+d$.
Suppose $b+c=3$. This implies $2b+1=3 \Longrightarrow b=1$. Hence $a=0$. And $a+d=4$ implies $0+d=4 \Longrightarrow d=4$.
Suppose $b+c=4$. This implies $2b+1=4 \Longrightarrow b=1.5$. Hence $a=-0.5$ and $a+d=3$ implies $-0.5+d=3 \Longrightarrow d=3.5$.

Qn14. Let $< DEF = x, < AED = p, < CEF = q$. Then $180^\circ = p+q+x$.



$$\begin{aligned} 180 &= \frac{180-A}{2} + \frac{180-C}{2} + x \\ 180 &= 180 + x - \frac{(A+C)}{2} \\ 180 &= 180 + x - 45 \\ x &= 45^\circ \end{aligned}$$

Qn15. (a) Let x be the middle integer. Then the other two integers are $x-1$ and $x+1$. Therefore $(x-1)(x)(x+1)+x = x[(x-1)(x+1)+1] = x(x^2-1+1) = x^3$ which is a cube as required.

(b) Suppose x and y are the grades Brenda obtained in the other two courses, then her overall average for the full year was

$$\begin{aligned} \frac{5(75)+4(80)+90+x+y}{12} &= 80 \\ x+y = 960-(5(75)+4(80)+90) &= 175. \\ x+y &= 175. \end{aligned}$$

The highest percentage grade she could obtain in either course is 100%.
Suppose $x = 100\%$, then $y = 175 - 100 = 75\%$ which is the minimum possible mark.

NMC 2016 0-LEVEL PAPER 2 QUESTIONS

SECTION A

Qn1. List all ordered triples (a, b, c) of positive integers which satisfy $a + \frac{b}{c} = 11$ and $b + \frac{a}{c} = 14$.

Qn2. In triangle $ABC, AB = 3, BC = 4$, and $AC = 6$. If BC is extended through C to D so that $CD = BC$, what is AD?

Qn3. Ampeirwe's Convenience Centre buys a calculator for p shillings (where $p > 0$), raises its price by $n\%$, then reduces this new price by 20%. If the final price is 20% higher than p, what is the value of n?

Qn4. A function f is defined so that if n is an odd integer, then $f(n) = n - 1$ and if n is an even integer, then $f(n) = n^2 - 1$. For example, if $n = 15$, then $f(n) = 14$ and if $n = -6$, then $f(n) = 35$, since 15 is an odd integer and -6 is an even integer. Determine all integers n for which $f(f(n)) = 3$.

Qn5. A student takes three exams. The second has twice as many questions as the first, and the third has three times as many questions as the first. The student answers exactly 75% of the questions correctly on the first exam, exactly 81% on the second exam, and exactly 85% on the third exam. Out of all the questions on the three exams, what percentage did he answer wrongly?

Qn6. The price of a ticket to attend a football match was Shs.10, 000 but due to a sponsorship deal by a company to pay up a certain amount of the ticket price for each person attending the match it was reduced. The number of spectators increased by 50%, but the amount of money received only increased by 20%. How much was the new ticket price after reduction?

Qn7. A large container, labeled R, is partially filled with 4 litres of red paint. Another large container, labeled W, is partially filled with 5 litres of white paint. A small empty can is completely filled with red paint taken from R, and the contents of the can then emptied into W. After thorough mixing of the contents of W, the can is completely filled with some of this mixture from W, and the contents of the can then emptied into R. The ratio of red paint to white in R is now 3 : 1. What is the size of the can, in litres?

Qn8. For what non-zero value of the radius, r, does the volume of a sphere equal its surface area?

Qn9. What is the tens digit of the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14$, and 15?

Qn10. Determine all values of x for which $\left(2 \cdot 4^{x^2-3x}\right)^2 = 2^{x-1}$.

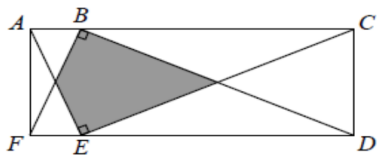
SECTION B

Qn11. My cat keeps to himself most of the time. I only heard him meow, hiss, and buzz on one out of the last 23 days. But I did hear him make at least one of these sounds each day. I heard him hiss on 6 days, buzz on 12 days, and meow on 15 days. On 2 days, I heard him meow and hiss but not buzz, and on 2 days, I heard him buzz and hiss but not meow. On how many days did I hear him meow and buzz but not hiss?

Qn12. Nankya and Kintu are driving on a circular track with radius 25 km. Nankya leaves the starting line first, going clockwise around the track at a speed of 80 km/h. Fifteen minutes after Nankya starts, Kintu leaves the same starting line, going counter- clockwise around the track at a speed of 100 km/h. For how many hours will Kintu have been driving when the two of them pass each other for the fourth time?

Qn13. In a psychology experiment, an image of a cat or an image of a dog is flashed briefly onto a screen and then Apio is asked to guess whether the image showed a cat or a dog. This process is repeated a large number of times with an equal number of images of cats and images of dogs shown. If Apio is correct 95% of the time when she guesses "dog" and 90% of the time when she guesses "cat", determine the ratio of the number of times she *guessed* "dog" to the number of times she *guessed* "cat".

Qn14. In the diagram, $ACDF$ is a rectangle with $AC = 200$ and $CD = 50$. Also, $\triangle FBD$ and $\triangle AEC$ are congruent triangles which are right-angled at B and E, respectively. What is the area of the shaded region?



Qn15. A formula for the sum of the first n positive integers is $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

(a) What is the sum of the first 200 positive integers, $1 + 2 + 3 + \cdots + 198 + 199 + 200$?

(b) Calculate the sum of the 50 consecutive integers beginning at 151, that is,

$$151 + 152 + 153 + \cdots + 198 + 199 + 200.$$

(c) Starting with the sum of the first 1000 positive integers, $1 + 2 + 3 + \cdots + 999 + 1000$, every third integer is removed to create the new sum

$$1 + 2 + 4 + 5 + 7 + 8 + 10 + 11 + \cdots + 998 + 1000.$$

Calculate this new sum.

NMC 2016 O-LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1.

$$a + \frac{b}{c} = 11 \Longrightarrow c = \frac{b}{11 - a} \tag{1}$$

$$b + \frac{a}{c} = 14 \Longrightarrow c = \frac{a}{14 - b} \tag{2}$$

Equating the two equations (1) and (2)

$$\frac{b}{11 - a} = \frac{a}{14 - b} \quad or \quad (11 - a)a = (14 - b)b$$

Clearly $a < 11$ and $b < 14$ since $a, b, \& \ c$ are positive integers.

By taking $a \in \{1, 2, \cdots, 10\}$, the solution set for the LHS is $\{10, 18, 24, 30\}$. Likewise, the solution set for the RHS is $\{13, 24, 33, 40, 48, 49\}$

The common solution is 24, which happens when $a = 8$ or $a = 3$ and $b = 12$ or $b = 2$.

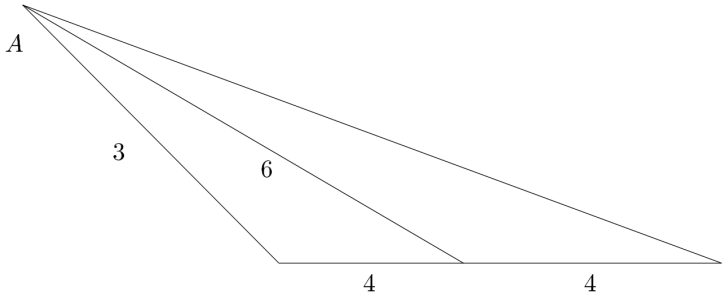
Case 1 For $a = 3, b = 2$ will make c in Eqn (2) non integer.

Case 2 For $a = 3, b = 12$ will make c in Eqn (2) non integer.

Case 3 For $a = 8, b = 2$ will make c in Eqn (2) non integer.

For $a = 8, b = 12$ gives an integer value of $c = 4$. Thus the solution set

$$(a, b, c) = (8, 12, 4)$$



Qn2.

By the Cosine rule

$$6^2 = 3^2 + 4^2 - 24\cos B \quad \Rightarrow \quad \cos B = -\frac{11}{24}$$

Similarly,

$$\begin{aligned} AD^2 &= 3^2 + 8^2 - 24\cos B \\ AD^2 &= 3^2 + 8^2 - 24\left(-\frac{11}{24}\right) \\ AD^2 &= 73 + 22 = 95 \\ &\Leftrightarrow \quad AD = \sqrt{95} \end{aligned}$$

Qn3. When the price of p shillings is raised by $n\%$, the price is multiplied by $1 + \frac{n}{100}$

When the new price is reduced by 20%, the new price is multiplied by $1 - \frac{20}{100} = \frac{80}{100}$ Therefore, after these two price adjustments, the price is

$$p\left(1 + \frac{n}{100}\right)\left(\frac{80}{100}\right)$$

We are told that this final price is 20% higher than p, and so the final price equals

$$p\left(1 + \frac{20}{100}\right) \quad \text{or} \quad p\left(\frac{120}{100}\right)$$

In other words,

$$p\left(1 + \frac{n}{100}\right)\left(\frac{80}{100}\right) = p\left(\frac{120}{100}\right)$$

Simplifying and using the fact that $p \neq 0$, we obtain

$$80\left(1+\frac{n}{100}\right)=120$$

Thus,

$$1+\frac{n}{100}=\frac{120}{80}=\frac{3}{2}=\frac{150}{100}$$

and so

$$\frac{n}{100}=\frac{50}{100} \quad \text{or} \quad n=50$$

Qn4. *Solution 1*

Let $m=f(n)$. The equation $f(f(n))=3$ becomes $f(m)=3$.

Suppose that $f(m)=3$ and m is odd. By definition, we have $f(m)=m-1=3$ and so $m=4$, which is not odd, so this case cannot happen.

Suppose that $f(m)=3$ and m is even. By definition, we have $f(m)=m^2-1=3$ and so $m^2=4$ or $m=\pm 2$, each of which is even.
Therefore, if $f(f(n))=3$, then $f(n)=2$ or $f(n)=-2$.

Suppose that $f(n)=2$ or $f(n)=-2$ and n is odd. By definition, we have $n-1=2$ (giving $n=3$) or $n-1=-2$ (giving $n=-1$). Each of these resulting values of n is odd.

Suppose that $f(n)=2$ or $f(n)=-2$ and n is even. Then $n^2-1=2$ or $n^2-1=-2$ which give $n^2=3$ or $n^2=-1$, neither of which is possible if n is an integer. Thus, the integers n for which $f(f(n))=3$ are $n=3$ and $n=-1$. (We can check by substitution that each of these satisfies the original equation.)

Solution 2

We consider the cases of n even and n odd separately.

Suppose that n is even. Then n^2 is even and so $f(n)=n^2-1$ must be odd. Thus, $f(f(n))=f(n^2-1)=(n^2-1)-1=n^2-2$, since $f(m)=m-1$ when m is odd.
For n to be even and $f(f(n))=3$, we must have $n^2-2=3$ or $n^2=5$. There are no integer solutions to this equation, and so there are no solutions in this case.

Suppose that n is odd. Then $f(n)=n-1$ must be even. Thus, $f(f(n))=f(n-1)=(n-1)^2-1=n^2-2n+1-1=n^2-2n$. For n to be odd and $f(f(n))=3$, we must have $n^2-2n=3$ or $n^2-2n-3=0$. Factoring, we obtain $(n-3)(n+1)=0$ and so $n=3$ or $n=-1$, both of which are odd.

Thus, the integers n for which $f(f(n))=3$ are $n=3$ and $n=-1$. (We can check by substitution that each of these satisfies the original equation.)

Qn5. Let the number of equations in the first exam be n, then the student answered correctly

$$\begin{aligned} & 0.75n+0.81\times 2n+0.85\times 3n \quad \text{out of} \quad (n+2n+3n) \\ \Rightarrow \quad & 0.75n+0.81\times 2n+0.85\times 3n \quad \text{out of} \quad 6n \end{aligned}$$

Those answered correctly

$$\frac{0.75n + 0.81 \times 2n + 0.85 \times 3n}{6n} = \frac{0.75 + 0.81 \times 2 + 0.85 \times 3}{6} = 0.82 = 82\%$$

Thus 82% answered correctly, and $100 - 82 = 18\%$ answered wrongly.

Qn6. If the new price ticket is x and the original number of spectators was N, then

$$1.5 \times Nx = 1.2 \times 10,000 \times N \quad \text{the increase in revenue by } 2\%$$

$$\Rightarrow \quad x = \frac{1.2(10,000N)}{1.5N} = 8,000$$

is the new ticket price.

Qn7. Let x denote the size of the can. After the first transfer, container R has

$$4 - x \text{ Red}$$

And container W has

$$x \text{ Red} \quad , \quad 5 \text{ White}$$

After the second transfer, R has

$$\left((4-x) + x\frac{x}{5+x}\right) \text{ Red} \quad , \quad x\frac{x}{5+x} \text{ White}$$

Thus using the ratio $1:3$

$$\begin{aligned} (4-x)(5+x) + x^2 &= 3(5x) \\ 20 &= 16x \end{aligned}$$

$$\Rightarrow \quad x = \frac{20}{16} = \frac{5}{4} \text{ litres}$$

Qn8. For the volume of a sphere equalling its surface area $\frac{4}{3}\pi r^3 = 4\pi r^2$.

$$\begin{aligned} \frac{4}{3}\pi r^3 - 4\pi r^2 &= 0 \\ 4\pi r^2\left(\frac{1}{3}r - 1\right) &= 0 \end{aligned}$$

Thus the non zero r value is $r = 3$.

Qn9. Among the list $10, 11, 12, 13, 14, 15$, the integers 11 and 13 are prime.
Also, $10 = 2 \times 5$ and $12 = 2 \times 2 \times 3$ and $14 = 2 \times 7$ and $15 = 3 \times 5$.
For an integer N to be divisible by each of these six integers, N must include at least two factors of 2 and one factor each of $3, 5, 7, 11, 13$.

Note that

$$2^2 \times 3 \times 5 \times 7 \times 11 \times 13 = 60060.$$

(This is the least common multiple of $10, 11, 12, 13, 14, 15$.)

To find the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14, 15$, we can find the smallest six-digit positive integer that is a multiple of 60 060.

Note that $1 \times 60\ 060 = 60\ 060$ and that $2 \times 60\ 060 = 120\ 120$. Therefore, the smallest six-digit positive integer that is divisible by each of $10, 11, 12, 13, 14, 15$ is 120 120. Therefore, the tens digit of this number is 2.

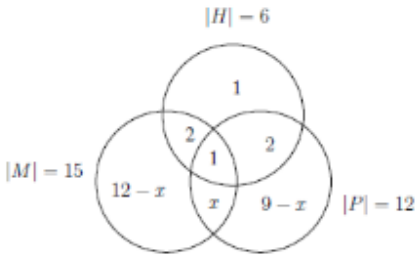
Qn10. Using exponent laws, we obtain the following equivalent equations:

$$\begin{aligned}
\left(2 \cdot 4^{x^2-3x}\right)^2 &= 2^{x-1} \\
2^2 \cdot 4^{2(x^2-3x)} &= 2^{x-1} \\
2^2 \cdot (2^2)^{2x^2-6x} &= 2^{x-1} \\
2^2 \cdot 2^{2(2x^2-6x)} &= 2^{x-1} \\
\frac{2^2 \cdot 2^{4x^2-12x}}{2^{x-1}} &= 1 \\
2^{2+4x^2-12x-(x-1)} &= 1 \\
2^{4x^2-13x+3} &= 1
\end{aligned}$$

Since $2^0 = 1$, this last equation is true exactly when $4x^2 - 13x + 3 = 0$ or $(4x-1)(x-3) = 0$. Therefore, $x = \frac{1}{4}$ or $x = 3$. We can check by substitution that each of these values of x satisfies the original equation.

SECTION B

Qn11. Let M denote the days on which the cat meowed, H the days on which he hissed, and P the days on which he buzzed. Given the data, the following Venn diagram represents the situation, where the expressions inside the circles are the cardinalities of disjoint sections:



We are looking for x. The labeled sections of the Venn diagram are disjoint sets, so they should add to 23. That is,

$$23 = 1 + 1 + 2 + 2 + x + (9 - x) + (12 - x) = 27 - x$$

Therefore, $x = 4$.

Qn12. Since the track is circular with radius 25 km, then its circumference is $2\pi(25) = 50\pi$ km.

In the 15 minutes that Nankya drives at 80 km/h, he drives a distance of $\frac{1}{4}(80) = 20$ km (because 15 minutes is one-quarter of an hour).

When Kintu starts driving, she drives in the opposite direction to Nankya. Suppose that Nankya and Kintu meet for the first time after Kintu has been driving for t hours. During this time, Kintu drives at 100 km/h, and so drives $100t$ km. During this time, Nankya drives at 80 km/h, and so drives $80t$ km.

Since they start $50\pi - 20$ km apart along the track (the entire circumference minus the 20 km that Nankya drove initially), then the sum of the distances that they travel is $50\pi - 20$ km. Therefore, $100t + 80t = 50\pi - 20$ and so $180t = 50\pi - 20$ or $t = \frac{5\pi-2}{18}$.

Suppose that Nankya and Kintu meet for the next time after an additional T hours. During this time, Kintu drives $100T$ km and Nankya drives $80T$ km. In this case, the sum of the distances that they drive is the complete circumference of the track, or 50π km. Thus, $180T = 50\pi$ or $T = 5\pi 18$.

The length of time between the first and second meetings will be the same as the amount of time between the second and third, and between the third and fourth meetings. Therefore, the total time that Kintu has been driving when she and Nankya meet for the fourth time will be

$$t + 3T = \frac{5\pi - 2}{18} + 3 \cdot \frac{5\pi}{18} = \frac{20\pi - 2}{18} = \frac{10\pi - 1}{9} \quad \text{hours}$$

Qn13. Suppose that Opio guesses "cat" c times and guesses "dog" d times.

When she guesses "dog", she is correct 95% of the time.
When she guesses "cat", she is correct 90% of the time.

Thus, when she guesses "cat", she is shown $0.9c$ images of cats and so $c - 0.9c = 0.1c$ images of dogs.

Thus, when she guesses "dog", she is shown $0.95d$ images of dogs and so $d - 0.95d = 0.05d$ images of cats.

(We assume that c and d have the property that $0.9c$ and $0.95d$ are integers.) Therefore, the total number of images of cats that she is shown is $0.9c + 0.05d$ and the total number of images of dogs that she is shown is $0.1c + 0.95d$.

But the number of images of cats equals the number of images of dogs.
Thus, $0.9c + 0.05d = 0.1c + 0.95d$, which gives $0.8c = 0.9d$ or $\frac{d}{c} = \frac{0.8}{0.9}$

Therefore, the ratio of the number of times that she guessed "dog" to the the number of times that she guessed "cat" is $8 : 9$.

Qn14. Since $\triangle FBD$ is congruent to $\triangle AEC$, then $FB = AE$. Since $\triangle FAB$ and $\triangle AFE$ are each right-angled, share a common side AF and have equal hypotenuses ($FB = AE$), then these triangles are congruent, and so $AB = FE$.

Now $BAFE$ has two right angles at A and F (so AB and FE are parallel) and has equal sides $AB = FE$ so must be a rectangle. This means that $BCDE$ is also a rectangle.

Now the diagonals of a rectangle partition it into four triangles of equal area. (Diagonal AE of the rectangle splits the rectangle into two congruent triangles, which have equal area. The diagonals bisect

each other, so the four smaller triangles all have equal area.)

Since $\frac{1}{4}$ of rectangle $ABEF$ is shaded and $\frac{1}{4}$ of rectangle $BCDE$ is shaded, then $\frac{1}{4}$ of the total area is shaded. (If the area of $ABEF$ is x and the area of $BCDE$ is y, then the total shaded area is $\frac{1}{4}x + \frac{1}{4}y$, which is $\frac{1}{4}$ of the total area $x + y$.)

Since $AC = 200$ and $CD = 50$, then the area of rectangle $ACDF$ is $200(50) = 10000$, so the total shaded area is $\frac{1}{4}(10000) = 2500$.

Qn15. (a) Using the formula given, the sum of the fist 200 positive integers

$$1 + 2 + 3 + \cdots + 198 + 199 + 200 \quad = \quad \frac{200(200 + 1)}{2} = 100(201) = 20100$$

(b) The sum of the fist 200 positive integers is equal to the sum of the fist 150 positive integers added to the sum of the 50 consecutive integers $151 + 152 + 153 + \cdots + 198 + 199 + 200$.

That is,

$$1 + 2 + \cdots + 198 + 199 + 200 = (1 + 2 + \cdots + 148 + 149 + 150) + (151 + 152 + \cdots + 198 + 199 + 200)$$

Therefore, the sum of the 50 consecutive integers $151 + 152 + 153 + \cdots + 198 + 199 + 200$ is equal to the difference between the sum of the fist 200 positive integers and the sum of the fist 150 positive integers. From part (15(15)a), the sum of the fist 200 positive integers is 20100.

Using the formula, the sum of the fist 150 positive integers

$$1 + 2 + 3 + \cdots + 148 + 149 + 150 \quad = \quad \frac{150(150 + 1)}{2} = 10075(151) = 11325$$

Therefore, the sum of the 50 consecutive integers beginning at 151,

$$\begin{aligned} 151 + 152 + \cdots + 199 + 200 &= (1 + 2 + \cdots + 199 + 200) - (1 + 2 + \cdots + 149 + 150) \\ &= 20100 - 11325 \\ &= 8775 \end{aligned}$$

(c) Let the required sum, $1 + 2 + 4 + 5 + 7 + 8 + 10 + 11 + \cdots + 998 + 1000$, be S.

Let the sum of the first 333 positive multiples of $3, 3 + 6 + 9 + 12 + \cdots + 996 + 999$, be T.

The sum of the first 1000 positive integers is equal to $S + T$. That is,

$$1 + 2 + \cdots + 998 + 999 + 1000 = (1 + 2 + 4 + 5 + 7 + \cdots + 998 + 1000) + (3 + 6 + \cdots + 996 + 999)$$

Therefore, the required sum S is equal to the difference between the sum of the first 1000 positive integers and T.

Using the formula, the sum of the first 1000 positive integers

$$1 + 2 + 3 + \cdots + 998 + 999 + 1000 \quad = \quad \frac{1000(1000 + 1)}{2} = 500(1001) = 500500$$

We may determine the sum T by first removing the common factor 3 from each term and then using the formula.

$$\begin{aligned} 3+6+9+\cdots+993+996+999 &= 3(1+2+3+\cdots+331+332+333) \\ &= 3\left(\frac{3333(333+1)}{2}\right) \\ &= 3(333\times 167) \\ &= 166833 \end{aligned}$$

Finally, the required sum S is $500500 - 166833 = 333667$.